MIT Art Design and Technology University's
MIT School of Computing, Pune
Department of Computer Science and Engineering
BTech Third Year **A.Y.2025-26**

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Artificial Intelligence and Machine Learning Lab

Enrolment No: ____________________ **Division:** _______________

Roll No: _______________ **Name:** _________________________________

Experiment No.: 06

Title: To study working of decision tree algorithm and visualize the results.

Theory:

The **ID3 (Iterative Dichotomiser 3)** algorithm is a fundamental decision tree algorithm used in classification tasks. It builds a decision tree by recursively selecting the **attribute that gives the highest information gain** at each node.

Key Concepts:

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1. **Decision Tree**:
   A tree-like model where each internal node represents a test on an attribute, each branch represents an outcome, and each leaf node represents a class label.
2. **Entropy (H)**:
   A measure of impurity or randomness in the data.
   Entropy(S)=−Σpilog2piEntropy(S) = -\sum p_i \log_2 p_iEntropy(S)=−Σpilog2pi
3. **Information Gain (IG)**:
   Reduction in entropy achieved by splitting the dataset on an attribute.
   Gain(S,A)=Entropy(S)−Σ(|Sv|/|S|⋅Entropy(Sv))Gain(S, A) = Entropy(S) - \sum \left(\frac{|S_v|}{|S|} \cdot Entropy(S_v) \right)Gain(S,A)=Entropy(S)−Σ(|S||Sv|⋅Entropy(Sv))
4. **Algorithm Flow (ID3)**:
   - Calculate entropy of the dataset
   - For each attribute, compute information gain
   - Choose attribute with highest gain as node
   - Repeat recursively for each branch until:
     - All attributes are used, or
     - All samples are classified
5. **Applications**:
   - Weather prediction
   - Loan eligibility
   - Medical diagnosis

- Email spam filtering

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Pseudo code or Program

```
# Import necessary libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

# Load the Iris dataset
iris = load_iris()
X = iris.data  # Features
y = iris.target  # Labels

# Split the dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)

# Create the Decision Tree Classifier with 'entropy' (ID3 logic)
clf = DecisionTreeClassifier(criterion='entropy', random_state=42)
clf.fit(X_train, y_train)

# Predict on test data
y_pred = clf.predict(X_test)

# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred,
target_names=iris.target_names))

# Visualize the tree
plt.figure(figsize=(12,8))
```

	plot_tree(clf, filled=True, feature_names=iris.feature_names, class_names=iris.target_names) plt.title("Decision Tree (ID3-style using Entropy) on Iris Dataset") plt.show()
Outputs	(Students can attach extra pages if necessary)

	Artificial Intelligence and Machine Learning Lab (Students can attach extra pages if necessary)
Exercise for Practice	
Q1.	What is the principle behind the ID3 algorithm, and how does it decide the best attribute to split?
Ans:	
Q2.	Define entropy and information gain with formulas. Why is information gain important in ID3?
Ans	
Q3	Consider a dataset with 9 positive and 5 negative instances. Calculate the entropy of the dataset.
Ans	
Q4.	What are the limitations of the ID3 algorithm? How do they affect real-world datasets?
Ans	

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Q5	Describe how the decision tree built using ID3 can be used to classify a new test sample. Give an example.
Ans	
Conclusion	Thus, we have successfully Studied decision treeClassifier and visualized it.

Date of Submission:	
Marks out of 10	**Name and Sign of Subject Teacher**
Remark:	